Worksheet 8.3

Science Quest 10: pages 311–12

Acceleration

Student: ..... Class: .....

The velocities of two objects were recorded over ten second time periods, as shown in the examples below.

Example: Gaining and losing speed

Time (s)	0	1	2	3	4	5	6	7	8	9	10
Velocity (m/s)	0	2	4	6	8	10	12	11	10	9	8

1. (a) Use a spreadsheet application such as Excel to plot a line graph of this data.

(b) Format your graph. Ensure that the graph has a title and that each axis has a title and appropriate units. Print a copy of this graph and attach it to this worksheet.

Use the printed graph to answer the following questions.

(c) Explain the shape of the graph.

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(d) At the eight second mark the object's speed begins to decrease. If the object's speed continues to decline at the same rate after ten seconds, at what time will its velocity be zero?

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Example: Falling down

Time (s)	0	1	2	3	4	5	6	7	8	9	10
Velocity (m/s)	0	9.8	19.6	29.4	39.2		58.8		78.4		98.0

2. (a) Examine the data table and determine the pattern in the data. Use this pattern to complete the table.

(b) Use a spreadsheet application such as Excel to plot a line graph of this data.

(c) Format your graph. Ensure that the graph has a title and that each axis has a title and appropriate units. Print a copy of this graph and attach it to this worksheet.

Use the printed graph to answer the following questions.

(d) Explain how this graph shows that falling due to gravity is an example of uniform acceleration.

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(e) The slope of the graph is equal to the acceleration due to gravity. Determine the slope of the graph in m/s^2.

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